

END TERM EXAMINATION**FOURTH SEMESTER [B.TECH] JUNE 2024****Paper Code: BS-202****Subject: Probability, Statistics &
Linear Programming****Time: 3 Hours****Maximum Marks: 75**

**Note: Attempt five questions in all including Q.No1 which is compulsory.
Select one question from each unit. Assume missing data if any.
(Scientific Calculators are allowed).**

Q1. Attempt all Questions:-**3**

a) A batch of 25 injection-molded parts contains 5 that have suffered excessive shrinkage. If two parts are selected at random and without replacement, what is the probability that the second part selected is one with excessive shrinkage?

b) State the Central Limit Theorem.

2

c) Find the moment-generating function of a random variable X following an exponential distribution.

3

d) Explain the general procedure for Hypothesis Testing.

2

e) Solve the following LPP graphically.

5

$$\begin{aligned} \text{Min } Z &= -3x + 2y \\ \text{subject to} \\ x + y &\leq 5 \\ 4 \leq x \leq 6, 6 \leq y \leq 1 \end{aligned}$$

UNIT I

Q2. a) Let the random variable X denote the current measured in a thin copper wire in milliamperes. Assume that the range of X is $[0, 20 \text{ mA}]$, and assume that the probability density function of X is $f(x) = 0.05, 0 \leq x \leq 20$. What is the probability that a measurement of current is between 5 and 10 milliamperes? Also, determine the mean and standard deviation of X . [7+8=15]

b) Customers are used to evaluate preliminary product designs. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and 25% have been poor products.

i. If a new design attains a good review, then what is the probability that it will be a highly successful product?

[-2]

- ii. If a product does not attain a good review, what is the probability that it will be a highly successful product?

Q3. a) A particularly long traffic light on your morning commute is green 20% of the time that you approach it. Assume that each morning represents an independent trial.

[7+3+5=15]

- i. Over five mornings, what is the probability that the light is green on exactly one day?
 ii. Over 20 mornings, what is the probability that the light is green on exactly four days?
 iii. Over 20 mornings, what is the probability that the light is green on more than four days?

b) Suppose the current measurements in a strip of wire are assumed to follow a normal distribution with a mean of 10 milliamperes and a variance of 4 (milliamperes)². What is the probability that a measurement will exceed 13 milliamperes?

c) Define Erlang Random Variable with its density function. Errors caused by contamination on optical disks occur at the rate of one error every 10^5 bits. Assume the errors follow a Poisson distribution. What is the probability that there are three or more errors within 10^5 bits?

UNIT II

Q4. a) Determine the value of c , and the covariance and correlation for the joint probability density function $f_{XY}(x, y) = c$ over the range $0 < x < 5, y > 0$, and $x - 1 < y < x + 1$.

[7+8=15]

b) A sample of the percentage mole conversion of naphthalene to maleic anhydride follows: 4.2, 4.7, 4.7, 5.0, 3.8, 3.6, 3.0, 5.1, 3.1, 3.8, 4.8, 4.0, 5.2, 4.3, 2.8, 2.0, 2.8, 3.3, 4.8, 5.0.

- i. Calculate the sample mean.
 ii. Calculate the sample variance and standard deviation.
 iii. Construct a box plot of the data

Q5. Suppose that X is a random variable with mean μ and variance σ^2 . Let $X_1, X_2, \dots, X_n$ be a random sample of size n from the population represented by X . Show that the sample mean $\bar{X}$ and sample variance S^2 are unbiased estimators of μ and σ^2 , respectively

[15]

UNIT III

Q6. a) Aircrew escape systems are powered by a solid propellant. The burning rate of this propellant is an important product characteristic. Specifications require that the mean burning rate

[7+8=15]

[-3]

must be 50 centimeters per second. We know that the standard deviation of burning rate is $\sigma = 2$ centimeters per second. The experimenter decides to specify a type I error probability or significance level of $\alpha = 0.05$ and selects a random sample of $n = 25$ and obtains a sample average burning rate of $\bar{x} = 51.3$ centimeters per second. What conclusions should be drawn?

b) Assume that the two variables, compressive strength (x) and intrinsic permeability (y) of various concrete mixes and cures are related according to the simple linear regression model. Using the following data, calculate the least squares estimates of the slope and intercept.

$$n = 14, \sum y_i = 572, \sum y_i^2 = 23530, \sum x_i = 43 \\ \sum x_i^2 = 157.42, \sum x_i y_i = 1697.80$$

Use the equation of the fitted line to predict what permeability would be observed when the compressive strength is 4.3

Q7. a) A random sample of 500 registered voters in Phoenix is asked if they favor the use of oxygenated fuels year-round to reduce air pollution. If more than 315 voters respond positively, we will conclude that at least 60% of the voters favor the use of these fuels.

[7+8=15]

- i. Find the probability of type I error if exactly 60% of the voters favor the use of these fuels.
 ii. What is the type II error probability if 75% of the voters favor this action?

b) Obtain a regression plane by using multiple linear regression to fit the data given below

x	1	2	3	4
y	0	1	2	3
z	12	18	24	30

UNIT IV

Q8. a) A head of department in a college has the problem of assigning courses to teachers with a view to maximize educational quality in his department. He has available to him one professor, two associate professors, and one teaching assistant (TA). Four courses must be offered. After appropriate evaluation, he has arrived at the following relative ratings (100 = best rating) regarding the ability of each instructor to teach each of the four courses. How should he assign his staff to the courses in order to realize his

[7+8=15]

objective?

	Course 1	Course 2	Course 3	Course 4
Prof 1	60	40	60	70
Prof 2	20	60	50	70
Prof 3	20	30	40	60
TA	30	10	20	40

b) Solve the following LPP using Simplex algorithm.

$$\text{Max } Z = -2x + 3y$$

subject to

$$x \leq 5$$

$$2x - 3y \leq 6$$

$$x, y \geq 0$$

Q9. a) Find the initial basic feasible solution of the following transportation problem using Vogel's Approximation method. [7+8=15]

	I	II	III	IV	Supply
A	21	16	25	13	11
B	17	18	14	23	13
C	32	27	18	41	19
Demand	6	10	12	15	

b) Three food products are available at costs of Rs. 10, Rs. 36 and Rs. 24 per unit, respectively. They contain 1,000, 4,000 and 2,000 calories per unit, respectively and 200, 900 and 500 protein units per unit, respectively. It is required to find the minimum-cost diet containing at least 20,000 calories and 3,000 units of protein. Formulate and solve the given problem as an LP problem. Write the dual and using the final simplex table of primal problem, write the solution to the dual problem.
